WORKBOOK 02 / 08 · MULTIMEDIA 12

Build a Caveman you can pose

Parts, symbols, pivots, hierarchy and readable silhouettes

ACT THROUGH THE WHOLE BODY



Balanced Thinking Proud

Three full-body acting studies: balanced, thinking and proud.

Build: Assemble an articulated character and demonstrate intentional joint movement.

Keep: Caveman_02_character.fla + neutral, thinking and proud pose images

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

A clear silhouette communicates action before details. A joint pivot controls how a body part rotates.

A flattened PNG is a picture, not an articulated rig. Fourteen separate Graphic symbols allow posing. The supplied art uses five original component drawings, reused for separate arm and leg symbols. Reusing an image does not mean the symbol definitions are the same. Parenting is optional for this core manual-posing route.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Pose the Caveman carrying the stone close to his chest. Preserve balance, clear hands and a grounded foot.

What you will defend

Show a joint pivot, explain why it belongs there, and identify one silhouette improvement.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Choose and inspect the construction route

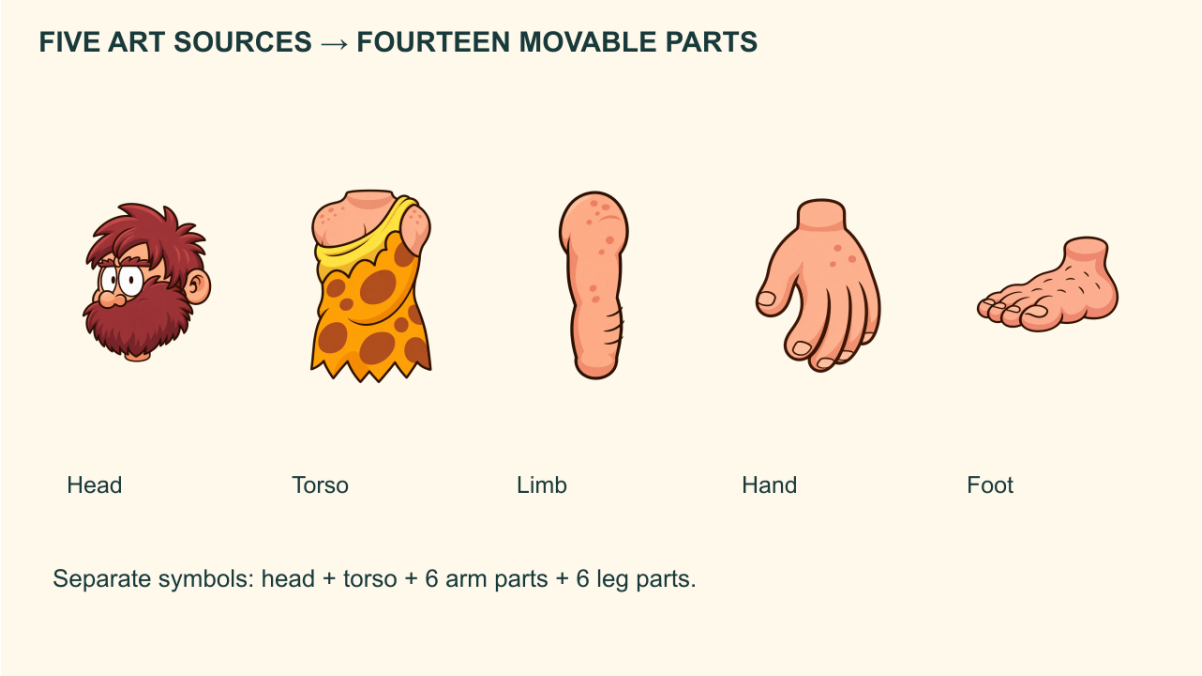
SELECT: Character artwork folder or the prepared character FLA.

TOOLS / ROUTE: File → Import → Import to Library; Window → Library.

1. Inspect the reference, head, torso, limb, hand and foot artwork. The original seated caveman_base.png is reserved for the counting close-up; it is not a walking body.

2. Choose the cutout route with supplied parts or draw your own matching parts with Brush/Pencil and fills. Both routes must create movable joints and the same story actions.

3. Use Far and Near to mean farther from and nearer to the viewer. Keep these names throughout the file.



Five original component artworks are reused in fourteen separate movable part definitions.

WHY · Construction choices affect what can move. Naming prevents the far and near limbs swapping identities halfway through a cycle.

CHECK / SUMMARY · You can identify the separate pieces and explain which picture is only a reference.

FIX · If you imported a single full-body image, retain it on a locked reference layer and build separate movable parts above it.

GLOSSARY · cutout animation: Animation using separately movable pieces.; silhouette: The readable outer shape of a pose.

DIFFERENT-OBJECT EXAMPLE · A paper bird needs a separate wing if the wing must flap independently.

STRETCH · Draw an alternate brow or hair tuft only if it can be placed without duplicating baked facial features.

02 · Create fourteen independent Graphic definitions

SELECT: One body part at a time; no background selected.

TOOLS / ROUTE: Modify → Convert to Symbol (F8); Window → Library → Duplicate.

1. Create Head and Torso. Create UpperArm_Far, Forearm_Far, Hand_Far, Thigh_Far, Shin_Far and Foot_Far; repeat the six names with _Near.

2. Use Graphic type. Duplicate a Library symbol before changing its internal artwork for another role. Give each stage instance its own layer of the same name.

3. When reusing the limb drawing, scale it inside each definition to suit upper arm, forearm, thigh or shin. Do not stretch the full character to fake bending.

WHY · Separate definitions let you revise one part without unintentionally replacing every limb.

CHECK / SUMMARY · Fourteen named Graphic definitions and fourteen part layers exist; selecting an arm does not select the whole Caveman.

FIX · If several pieces edit together, exit symbol editing and check whether you reused one definition. Duplicate and replace the appropriate instance.

GLOSSARY · definition: The reusable master stored in the Library.; instance: A placed use of a Library definition.

DIFFERENT-OBJECT EXAMPLE · Four wheel instances can share one wheel definition; a bent wheel needs a separate definition if only it should change.

STRETCH · Explain a situation where shared definitions are useful and one where they are dangerous.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Place pivots at the actual joints

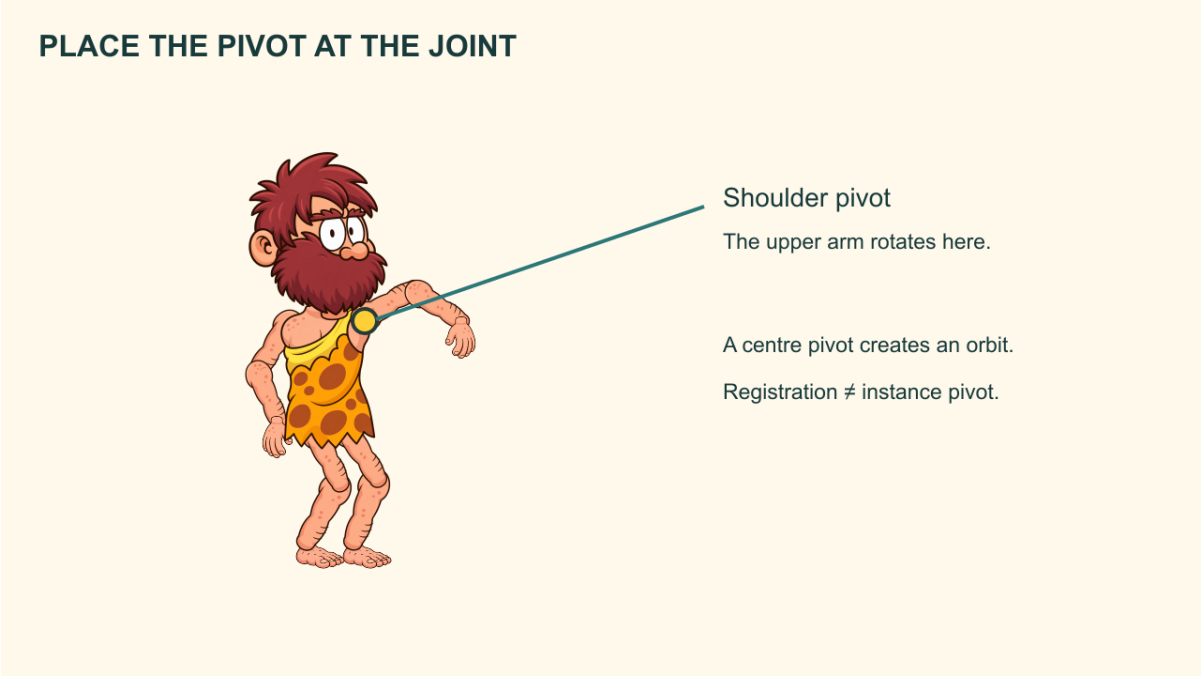
SELECT: An individual part instance with Free Transform (Q).

TOOLS / ROUTE: Free Transform → move the small white transformation point.

1. For each upper arm and thigh, move the transformation point to the shoulder or hip. Put the forearm pivot at the elbow, shin pivot at the knee, hand pivot at the wrist and foot pivot at the ankle.

2. Rotate a small amount, then undo. The joined end should remain in place. Test in both directions.

3. Distinguish the movable instance pivot from the symbol registration origin and from transparent PNG margins. Judge the joint by visible anatomy.



Joint-pivot mechanism diagram. The yellow point marks the shoulder; this is not a simulated Animate interface.

WHY · A centre pivot makes an arm orbit the shoulder. A joint pivot makes it rotate from the shoulder.

CHECK / SUMMARY · The test rotation leaves the joint attached. You can point to the pivot on the selected instance.

FIX · If the part swings away, undo and move the white point to the joint. Do not repair a pivot error by dragging the whole body after every rotation.

GLOSSARY · pivot: The transformation point used for an instance rotation.; registration: The symbol's internal coordinate origin.

DIFFERENT-OBJECT EXAMPLE · A door rotates from its hinge, not from the middle of the door panel.

STRETCH · Compare a hand waving from its wrist with an upper arm swinging from its shoulder.

04 · Assemble a neutral pose in depth order

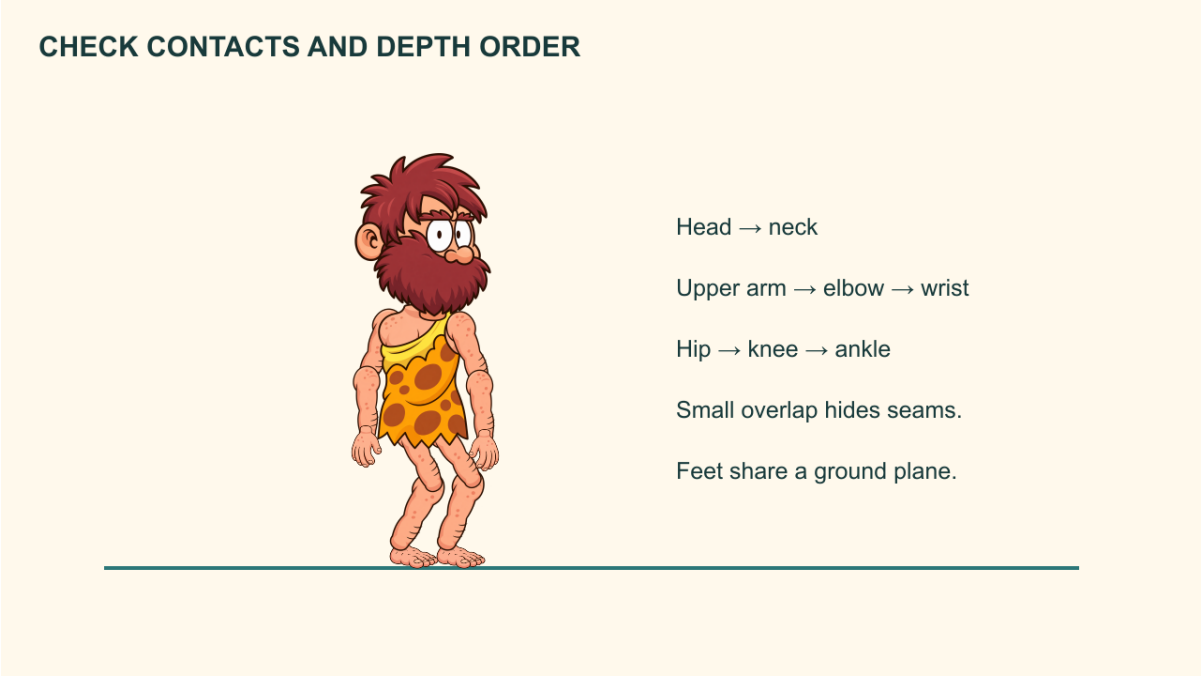
SELECT: The fourteen part layers at frame 1.

TOOLS / ROUTE: Timeline layer order; Selection and Free Transform.

1. Place feet on one ground line. Build shins to knees, thighs to hips, torso to hips, arms to shoulders and head to neck. Allow small joint overlaps so gaps do not open when posing.

2. Put far limbs behind the torso and near limbs in front where the pose requires it. Keep the head above the torso.

3. Select the complete assembly and convert it to a Graphic named Caveman_Action. Enter it, select the fourteen instances and use Modify → Timeline → Distribute to Layers if conversion put them on one layer; rename the resulting inner layers. Verify one part per inner layer, then inspect and restore the intended depth order; automatic distribution can follow creation order. Keep a neutral one-frame Library template.



Assembly study: connected joints, deliberate depth order and a shared ground line.

WHY · Layer order creates depth; overlap hides seams. A single action container lets the whole performance be staged later.

CHECK / SUMMARY · The neutral character has attached joints, a consistent head size and no floating feet.

FIX · If a limb disappears, check visibility, layer order and scale before importing another copy. If a dark halo appears, inspect the source edge on a light background.

GLOSSARY · overlap: Artwork covering a small part of another piece to hide a joint seam.; depth order: Which pieces appear in front of others.

DIFFERENT-OBJECT EXAMPLE · A puppet's sleeve overlaps its upper arm so the seam stays hidden while it bends.

STRETCH · Try a three-quarter pose without swapping the names Near and Far.

05 · Test a hierarchy without depending on it

SELECT: A duplicate test symbol, not the master neutral pose.

TOOLS / ROUTE: Timeline → Parenting view (Advanced Layers); or use the manual fallback.

1. Optional: enable the Parenting view, then parent a forearm layer to its upper arm and a hand to its forearm. Use the parent control or drag between parent circles as shown by your installed workspace.

2. Rotate the upper arm. Confirm whether both children follow. Undo and save the successful test separately. Parenting has not been certified by the supplied author test; stop this branch if the installed control differs.

3. Core fallback: leave parenting off. Rotate the upper arm, move the elbow end into place, rotate the forearm about its elbow, then place the hand at the wrist. Repeat for each designed pose.

WHY · A hierarchy can carry child transforms, but it does not choose a good pose, bend a knee intelligently or lock a foot to the ground.

CHECK / SUMMARY · Either you demonstrate the child-follow test or clearly identify that your file uses manual joint placement.

FIX · If a joint jumps after parenting, undo. Inspect pivots and the parent assignment in the duplicate; continue manually rather than breaking the master.

GLOSSARY · parenting: A relationship in which a child follows transformations of its parent.; inverse kinematics: A solving method that can calculate joint positions from an endpoint; ordinary parenting is not this.

DIFFERENT-OBJECT EXAMPLE · Moving a desk can carry a lamp placed on it, but does not automatically keep the lamp level in an animation hierarchy.

STRETCH · Build a one-chain parent test and show precisely what follows and what does not.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

06 · Design three readable acting poses

SELECT: Neutral template duplicated into a Pose_Test Graphic.

TOOLS / ROUTE: Insert Keyframe at frames 1, 13 and 25; Free Transform.

1. At frame 1 make a balanced neutral pose. At frame 13 make a thinking pose: head toward the goal, one hand near the beard, shoulders and hips still plausible.

2. At frame 25 make a proud pose: open chest, a deliberate hand gesture and a stable base. Rotate parts, reposition joints and inspect the line of action.

3. View the poses as silhouettes or squint at the Stage. Separate the hands from the torso when their meaning must be read.

WHY · Acting is communicated by the relationship of body parts, not just by a different mouth image.

CHECK / SUMMARY · A viewer can identify thinking and pride before you name the poses.

FIX · If two poses read the same, change the torso lean, head direction and negative space before adding small facial details.

GLOSSARY · line of action: A simple directional curve expressing the pose’s energy.; negative space: The clear spaces between parts of a silhouette.

DIFFERENT-OBJECT EXAMPLE · A person offering a gift extends the object into clear space; hiding it against the chest changes the message.

STRETCH · Make a doubtful proud pose by changing posture while keeping the same facial artwork.

07 · Repair the broken-joint example

SELECT: A copy of Pose_Test with one intentionally misplaced pivot.

TOOLS / ROUTE: Free Transform; timeline keys; Undo.

1. Move one elbow pivot to the centre of the forearm in the duplicate and rotate it. Capture the visible gap.

2. Restore the pivot at the elbow and reposition the wrist. Test the full range used by the thinking pose.

3. Save before/after evidence and explain the difference between the joint error and a layer-order error.

WHY · A useful repair demonstrates the cause of a failure, rather than hiding it behind another part.

CHECK / SUMMARY · The same rotation produces an attached joint after your repair.

FIX · If the gap remains, check that the pivot is correct on that keyframe instance and that the two visible joint ends overlap.

GLOSSARY · diagnosis: Identifying the cause of a visible problem.

DIFFERENT-OBJECT EXAMPLE · A wheel orbiting its axle is a pivot error; a wheel hidden behind the wrong panel is a depth-order error.

STRETCH · Check both the most bent and most extended poses for seams.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Pose the changed carrying condition

SELECT: A duplicate character action for your director version.

TOOLS / ROUTE: Save As; pose keys; Stage preview.

1. Create a carrying pose with the stone close to the chest. Use both hands if necessary, but keep the object visible.

2. Keep the supporting foot under the body's apparent weight. Show a front/three-quarter readability check at small size.

3. Save and reopen. Retain a neutral template and your three acting poses for the following workbooks.

WHY · A changed prop condition requires new joint relationships and balance decisions.

CHECK / SUMMARY · The stone appears held, hands remain attached and the pose differs intentionally from the guided examples.

FIX · If the stone floats, align fingers or palms around it and hold its position relative to the carrying hand.

GLOSSARY · balance: A pose's apparent ability to support its own weight.

DIFFERENT-OBJECT EXAMPLE · No worked carrying solution: use contact, silhouette and balance criteria.

STRETCH · Design a one-handed carry and explain the counterbalance in the torso.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

Original teacher-supplied Caveman PDF/PNG assets are preserved. New matching Caveman parts/backgrounds are AI-generated and identified in the asset register. Teaching diagrams are authored illustrations; genuine native captures are labelled separately. Supplied commercial course videos/FLAs are reference material and are not redistributed.

Publication/classroom status: see the teacher companion. This file does not certify the independent cold run, a natural voice performance, or live assessment/progress connections.